

3.1 FACTOR REGRESSIONS

We can estimate the risk-adjusted benchmark, or equivalently the mimicking portfolio, using factor regressions.

CAPM Benchmark

To illustrate, consider the grand master of value investing, Warren Buffett. Taking monthly returns on Berkshire Hathaway from 1990 to May 2012, I run the following CAPM regression:

$$r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + \varepsilon_{it}, \tag{10.9}$$

where ordinary least squares assumes that the residuals ε_{it} are independent of the market factor. Estimating the CAPM regression in equation (10.9) implicitly yields the CAPM-implied mimicking portfolio weights. The estimates for Berkshire Hathaway are

	Coefficient	T-stat
Alpha	0.72%	2.02
Beta	0.51	6.51
Adj R^2	0.14	

This implies that the CAPM benchmark consists of 0.49 in T-bills and 0.51 in the market portfolio, $r^{bmk} = 0.49r_f + 0.51r_m$, and that \$1 invested in Berkshire Hathaway is equivalent to

$$\begin{aligned} (1 - 0.51) &= \$0.49 \text{ in T-bills} \\ &+ \$0.51 \text{ in the market portfolio.} \end{aligned}$$

Relative to this benchmark portfolio, Buffett is adding

$$+ 0.72\% \text{ (alpha) per month.}$$

This is impressive performance! Buffett is generating an alpha of $0.0072 \times 12 = 8.6\%$ per year with a risk of approximately half that of the market ($\beta = 0.51$). The alpha estimated using a market portfolio benchmark in equation (10.9) is often called Jensen’s alpha, after Michael Jensen’s pioneering 1968 study of mutual fund performance. (No, mutual funds generally don’t beat the market; see chapter 16.)

The alpha is also statistically significant, with a high t -statistic above two. The cutoff level of two corresponds to the 95% confidence level, a magic threshold for statisticians. Buffett is special. Most factor regressions do *not* produce significant alpha estimates. The adjusted R^2 of the CAPM regression is 14%, which is also relatively high. For most stocks, CAPM regressions produce R^2 s of less than 10%.⁷

⁷ CAPM regressions tend to produce low R^2 s because idiosyncratic risk is large in stock returns. Economically, there are tremendous diversification benefits in creating portfolios. See chapter 3.